

# Crystal planes in cubic cells: how to see and slice them

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## 1. Basics: coordinates and atoms in the cube

I put the origin at one corner of the cube and measure every position in units of the lattice parameter  $a$ . The cube then runs from 0 to 1 along  $x$ ,  $y$  and  $z$ , and I write each atom as fractional coordinates  $(x, y, z)$ .

I draw these atoms in one cube (every ball a textbook picture shows):

| Structure         | Atom positions I draw                                             | Balls drawn | Atoms the cell owns            |
|-------------------|-------------------------------------------------------------------|-------------|--------------------------------|
| Simple cubic (SC) | 8 corners                                                         | 8           | $8 \times 1/8 = 1$             |
| BCC               | 8 corners + body center $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ | 9           | $1 + 1 = 2$                    |
| FCC               | 8 corners + 6 face centers                                        | 14          | $1 + 6 \times \frac{1}{2} = 4$ |

The FCC face centers sit at  $(\frac{1}{2}, \frac{1}{2}, 0)$ ,  $(\frac{1}{2}, \frac{1}{2}, 1)$ ,  $(\frac{1}{2}, 0, \frac{1}{2})$ ,  $(\frac{1}{2}, 1, \frac{1}{2})$ ,  $(0, \frac{1}{2}, \frac{1}{2})$  and  $(1, \frac{1}{2}, \frac{1}{2})$ .

Drawn versus owned: 8 cubes share each corner atom and 2 cubes share each face atom, so the FCC cell owns 4 atoms. When I draw, slice or project a cube, I always work with all 14 balls. "4 atoms" and "14 balls" are both right; they count different things.

## 2. Miller indices and the plane equation

I keep two things apart:  $h, k, l$  belong to the plane, and  $x, y, z$  belong to an atom. The Miller indices  $(hkl)$  are three fixed numbers that name a plane family. The coordinates  $(x, y, z)$  change from atom to atom.

How I get the equation:

1. The plane  $(hkl)$  cuts the  $x, y$  and  $z$  axes at  $1/h, 1/k$  and  $1/l$ . An index of 0 means the plane never cuts that axis (intercept at infinity).
2. Any flat plane that cuts the axes at  $p, q$  and  $r$  follows the intercept form  $x/p + y/q + z/r = 1$ .

3. I substitute  $p = 1/h$ ,  $q = 1/k$ ,  $r = 1/l$ .

Plane equation:  $h \cdot x + k \cdot y + l \cdot z = 1$

My checks:

- (111) cuts all three axes at 1, so I get  $x + y + z = 1$ . This is the yellow triangle through (1,0,0), (0,1,0) and (0,0,1).
- (001) never cuts x or y and cuts z at 1, so I get  $0 \cdot x + 0 \cdot y + 1 \cdot z = 1$ , which is  $z = 1$ : the top face of the cube.
- (110) cuts x and y at 1 and never cuts z, so I get  $x + y = 1$ : a vertical rectangle through the cube diagonal.

Negative index: I shift the origin to a neighboring corner, or I keep the minus sign in the formula. The sorting method in section 3 works the same either way.

### 3. Slices: sorting atoms onto parallel planes

Every plane parallel to (hkl) keeps the same left side and changes only the number on the right. I treat that right-hand number as a slice number.

Parallel slices:  $h \cdot x + k \cdot y + l \cdot z = s$ , where  $s = 0, \frac{1}{2}, 1, 1.5, 2$ , and so on

How I use it:

1. I take one atom's coordinates (x, y, z).
2. I compute  $s = hx + ky + lz$ .
3. Atoms with the same s share a slice. Atoms with different s sit on different, parallel slices.

My examples for (111), where  $s = x + y + z$ :

- (0,0,0) gives 0, so it sits on slice 0.
- ( $\frac{1}{2}, \frac{1}{2}, 0$ ) gives 1, so it sits on slice 1, the yellow triangle.
- (1,1,0) gives 2, so it sits on slice 2.

Special case (001):  $s = z$ , so the slice number is just the atom's height. I picture the slices as floors of a building.

In this note, "layer" and "slice" mean the same thing: one flat sheet of atoms parallel to the (hkl) plane.

## 4. How many slices, and how far apart

I never choose the slices; the atoms decide them. A slice exists only where an atom sits.

1. I compute  $s = hx + ky + lz$  for every ball in the cube.
2. I count the different values of  $s$ . That count gives the number of slices inside the cube.
3. I take the smallest gap  $\Delta s$  between neighboring values.
4. I convert that gap to a distance.

Spacing:  $d = \Delta s \times a / \sqrt{h^2 + k^2 + l^2}$

My FCC results:

| Plane | Values of $s$                | Slices in the cube | Smallest gap $\Delta s$ | Spacing $d$              | Balls per slice |
|-------|------------------------------|--------------------|-------------------------|--------------------------|-----------------|
| (001) | 0, $\frac{1}{2}$ , 1         | 3                  | $\frac{1}{2}$           | $a/2 = 0.500a$           | 5, 4, 5         |
| (110) | 0, $\frac{1}{2}$ , 1, 1.5, 2 | 5                  | $\frac{1}{2}$           | $a/(2\sqrt{2}) = 0.354a$ | 2, 2, 6, 2, 2   |
| (111) | 0, 1, 2, 3                   | 4                  | 1                       | $a/\sqrt{3} = 0.577a$    | 1, 6, 6, 1      |

These spacings match the FCC d-spacings I see in XRD:  $d(111) = a/\sqrt{3}$ ,  $d(200) = a/2$ ,  $d(220) = a/(2\sqrt{2})$ . This explains why FCC shows the (200) and (220) peaks instead of (100) and (110): the real slices sit twice as close as the (100) and (110) labels suggest.

Why I use the smallest gap: one cube does not hold every slice of the crystal. In BCC (111), the cube gives  $s = 0, 1, 1.5, 2, 3$ , so the gaps come out uneven. Body centers of neighboring cubes fill the missing values (0.5, 2.5). The smallest gap,  $\frac{1}{2}$ , gives the true spacing  $a/(2\sqrt{3}) = 0.289a$ .

Edge slices: the slice at the top of one cube repeats the slice at the bottom of the next. For FCC (001),  $z = 1$  repeats  $z = 0$ , so only two kinds of slice alternate: corner-type and middle.

## 5. Two ways to look at a plane

Every plane figure shows one of two pictures. I got confused before because I did not know which one I was looking at.

|             | Picture A: single slice              | Picture B: projection              |
|-------------|--------------------------------------|------------------------------------|
| What I keep | Only the atoms with one value of $s$ | All atoms in the cell, every slice |

|                 | Picture A: single slice                                                     | Picture B: projection                                                             |
|-----------------|-----------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| How I look      | Straight at the sheet                                                       | Straight down $[hkl]$ , through all slices                                        |
| Distances I see | True distances                                                              | True only within one slice; atoms from different slices look closer than they are |
| Overlaps        | None                                                                        | Atoms on one line of sight land on one spot                                       |
| Where I meet it | Planar density, textbook plane drawings, the top layer of a cleaved surface | Journal figure insets, "view along $[hkl]$ " images                               |

Viewing direction rule (cubic only): the direction  $[hkl]$  runs perpendicular to the plane  $(hkl)$ . To look straight at a  $(111)$  plane, I look along  $[111]$ , the body diagonal. In non-cubic cells this rule fails.

Which slice I pick for picture A: in a one-atom-per-lattice-point metal (Cu, Al, W), every slice of a family carries the same atom pattern, only shifted sideways. Any slice gives the same picture. By convention I draw the slice with  $s = 1$ , because it cuts the axes at  $1/h$ ,  $1/k$ ,  $1/l$  and fits inside one cube.

The sideways shift between slices is the stacking sequence. In FCC  $(111)$ , three shifted positions repeat: ABC stacking.

How I spot a projection: if a figure shows atoms that cannot all satisfy the same value of  $hx + ky + lz$ , I know it shows picture B.

## 6. Worked examples: FCC (001), (110), (111)

FCC (001):  $s = z$

| s             | Balls                                                                                                                                      | Count |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 0             | (0,0,0), (1,0,0), (0,1,0), (1,1,0), ( $\frac{1}{2}, \frac{1}{2}, 0$ )                                                                      | 5     |
| $\frac{1}{2}$ | ( $\frac{1}{2}, 0, \frac{1}{2}$ ), ( $\frac{1}{2}, 1, \frac{1}{2}$ ), ( $0, \frac{1}{2}, \frac{1}{2}$ ), ( $1, \frac{1}{2}, \frac{1}{2}$ ) | 4     |
| 1             | (0,0,1), (1,0,1), (0,1,1), (1,1,1), ( $\frac{1}{2}, \frac{1}{2}, 1$ )                                                                      | 5     |

- Picture A (slice  $z = 0$ ): I see an  $a \times a$  square with 4 corners and 1 center ball. The nearest-neighbor distance is  $a/\sqrt{2}$ , corner to center.
- Picture B (looking down  $z$ ): slices  $z = 0$  and  $z = 1$  fall exactly on each other. I see 9 spots: 4 corners, 1 center, and 4 edge midpoints from slice  $z = \frac{1}{2}$ . The  $z = \frac{1}{2}$  balls form a square of side  $a/\sqrt{2}$ , rotated  $45^\circ$ .

FCC (110):  $s = x + y$

| s             | Balls                                                                                                    | Count |
|---------------|----------------------------------------------------------------------------------------------------------|-------|
| 0             | (0,0,0), (0,0,1)                                                                                         | 2     |
| $\frac{1}{2}$ | ( $\frac{1}{2}, 0, \frac{1}{2}$ ), ( $0, \frac{1}{2}, \frac{1}{2}$ )                                     | 2     |
| 1             | (1,0,0), (0,1,0), (1,0,1), (0,1,1), ( $\frac{1}{2}, \frac{1}{2}, 0$ ), ( $\frac{1}{2}, \frac{1}{2}, 1$ ) | 6     |
| 1.5           | ( $1, \frac{1}{2}, \frac{1}{2}$ ), ( $\frac{1}{2}, 1, \frac{1}{2}$ )                                     | 2     |
| 2             | (1,1,0), (1,1,1)                                                                                         | 2     |

- Picture A (slice  $s = 1$ ): I see a rectangle  $a/\sqrt{2}$  wide and  $a$  tall, with balls at the 4 corners and at the middle of the top and bottom edges. Spacing along the long edge is  $a/\sqrt{2}$ .
- Picture B (looking down  $[110]$ ): I see 8 spots in an  $a/\sqrt{2} \times a$  rectangle. The bottom row has 3 spots, and its middle spot stacks 3 balls: (0,0,0), (1,1,0) and ( $\frac{1}{2}, \frac{1}{2}, 0$ ). The top row repeats this at  $z = 1$ . The middle row has 2 spots at  $\pm 0.354a$  from center, each holding 2 balls from slices  $\frac{1}{2}$  and 1.5.

FCC (111):  $s = x + y + z$ 

| s | Balls                                                                                                                              | Count |
|---|------------------------------------------------------------------------------------------------------------------------------------|-------|
| 0 | (0,0,0)                                                                                                                            | 1     |
| 1 | (1,0,0), (0,1,0), (0,0,1), ( $\frac{1}{2}, \frac{1}{2}, 0$ ), ( $\frac{1}{2}, 0, \frac{1}{2}$ ), ( $0, \frac{1}{2}, \frac{1}{2}$ ) | 6     |
| 2 | (1,1,0), (1,0,1), (0,1,1), ( $1, \frac{1}{2}, \frac{1}{2}$ ), ( $\frac{1}{2}, 1, \frac{1}{2}$ ), ( $\frac{1}{2}, \frac{1}{2}, 1$ ) | 6     |
| 3 | (1,1,1)                                                                                                                            | 1     |

- Picture A (slice  $s = 1$ ): I see an equilateral triangle of side  $a\sqrt{2}$ , with corner balls at its tips and face-center balls at the midpoints of its edges. The nearest-neighbor distance is  $a/\sqrt{2}$ .
- Picture B (looking down  $[111]$ , the body diagonal): the four slices stack into one image.
  - Slice 1 shows as the triangle at true size.
  - Slice 2 shows the same triangle turned  $180^\circ$ . Together they form a six-pointed star.
  - Slices 0 and 3 are single corners on my line of sight, so both land on the center spot.
  - Result: a hexagon (the cube's outline seen corner-on) with 6 tip spots at radius  $a\sqrt{2}/3 = 0.816a$ , 6 inner spots at  $a/\sqrt{6} = 0.408a$ , and 1 center spot holding 2 balls. 13 spots, 14 balls.

## The 2D repeat cell of a (111) slice

The triangle is only the part of the slice that the cube walls cut off. It cannot serve as a unit cell, because triangles do not tile a plane by translation alone. The smallest repeat unit is a rhombus with side  $a/\sqrt{2}$  and angles of  $60^\circ$  and  $120^\circ$ , holding 1 atom. My example:  $(1,0,0)$ ,  $(\frac{1}{2}, \frac{1}{2}, 0)$ ,  $(0, \frac{1}{2}, \frac{1}{2})$ ,  $(\frac{1}{2}, 0, \frac{1}{2})$ , all from slice 1.

My check: the triangle has area  $\sqrt{3}a^2/2$  and holds  $3 \times 1/6 + 3 \times 1/2 = 2$  atoms. The rhombus has area  $\sqrt{3}a^2/4$  and holds 1 atom. One triangle equals two rhombi.

## 7. Reading a plane figure someone shows me

My checklist:

1. I identify the structure (SC, BCC, FCC) and list the balls of one cube with their coordinates.
2. I compute  $s = hx + ky + lz$  for each ball and group the balls into slices.

3. I decide whether the figure shows picture A (one slice) or picture B (projection).  
Atoms from more than one slice mean picture B.
4. I count spots. For picture B, I compare spots with balls: missing balls hide behind others on the same line of sight.
5. I check distances. Within one slice, the nearest FCC distance is  $a/\sqrt{2}$ . Any labeled distance or drawn cell should join atoms of the same slice.
6. For a cleaved surface or a 2D cell, I expect a parallelogram (a rhombus for (111)), not the triangle.

### Case: Materials Studio cleave

- Cleave (111) builds a periodic slab, so its in-plane cell is the rhombus:  $a = b = a(\text{FCC})/\sqrt{2}$ ,  $\gamma = 120^\circ$  (or  $60^\circ$ ). It never shows the triangle.
- The window shows periodic images on the cell edges and may show several slices, so the ball count differs from one slice.
- To compare with a textbook projection, I use a single-element FCC conventional cell and view exactly along [111]. In VESTA, I set the projection vector  $u\ v\ w$  in the Orientation dialog.
- NaCl (rock salt) consists of two interpenetrating FCC lattices, so its pictures differ from single-element FCC.

## 8. Quick reference card

- Coordinates: I put the origin at a corner, use units of  $a$ , and let the cube run from 0 to 1.
- Balls I draw: SC 8, BCC 9, FCC 14. Atoms the cell owns: 1, 2, 4.
- Plane ( $hkl$ ): cuts the axes at  $1/h$ ,  $1/k$ ,  $1/l$ . Equation  $h \cdot x + k \cdot y + l \cdot z = 1$ .
- Slice number of an atom:  $s = hx + ky + lz$ . Same  $s$  means same slice.
- Number of slices in a cube: I count the different values of  $s$ .
- Spacing:  $d = (\text{smallest gap in } s) \times a / \sqrt{h^2 + k^2 + l^2}$ .
- To see ( $hkl$ ) face-on, I look along  $[hkl]$  (cubic only).
- Picture A shows one slice with true distances. Picture B stacks all slices, with overlaps.
- FCC: (001) has 3 slices,  $d = a/2$ ; (110) has 5 slices,  $d = 0.354a$ ; (111) has 4 slices,  $d = 0.577a$ .
- FCC (111) projection: a hexagon with 13 spots for 14 balls; a star of two triangles plus a doubled center.
- 2D cell of FCC (111): a rhombus, side  $a/\sqrt{2}$ ,  $60^\circ/120^\circ$ , 1 atom. The triangle is a cube cut, not a cell.